MULAN ZHOU

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RESEARCH INTERESTS

Empirical AI safety and language-model behavior: representation-level readouts and controlled interventions; agent intent monitoring; constraint availability, enforceability, and evaluation reliability.

EDUCATION

University of California, San Diego Expected June 2028 | GPA: 3.85/4.00

B.S. in Cognitive Science, Specialization in Machine Learning and Neural Computation; Minor in Computer Science

RESEARCH EXPERIENCE

Selective Rule Updating & Mechanistic Evaluation 2026–Present

- Designed and ran GPU-based studies on Qwen2.5-1.5B-Instruct, using a preregistered 2,592-record evaluation with held-out tests across 18 structural families.
- Analyzed hidden-state probes and activation interventions: rule applicability was highly decodable (AUROC 0.991), but did not meet the preregistered standard for a dominant causal mechanism.
- Evaluated Qwen2.5-14B-Instruct on 1,024 cases: 46.2% direct accuracy versus 89.6% with a separate free-reasoning scaffold; 400/400 when task operands were explicit.
- Tested whether controlled activation edits exposed the model's decision mechanism: the edits changed an internal readout, not final answers; readout contraction explained 74.3% of the apparent interaction.

Instruction Availability & Agent Constraints 2026

- Investigated two distinct agent-safety failure modes: source constraints omitted before a model decision, and downstream guards unable to represent or enforce the required constraint.
- Used pinned-stack replay and interface-level analysis of a financial-agent workflow to trace tool-output truncation, constraint delivery, and available pre-trade enforcement inputs.
- Separated evidence that a constraint reaches the decision point from evidence that the agent obeys it, motivating decision-point auditing rather than relying only on aggregate compliance scores.

Intent Drift Verification 2025–Present

- Co-developed Authority, Coherence, Scope, and Impact evaluation engines to compare multi-step agent behavior with structured authorized intent and assess drift severity.
- Built a semantic Coherence Engine with transformer-based natural-language inference and sentence embeddings; helped construct IDV-1000 (850 author-injected, 150 naturalistic traces).
- Measured 30% disagreement between two frontier LLM judges on tested naturalistic traces, reaching 82.4% in the reported instance-level entity-access subset.
- Achieved 93.9% F1 and 98.7% AUROC on the 850 author-injected traces, with 6.5× fewer false positives than the evaluated LLM-judge baselines under the reported setup.

PUBLICATIONS & MANUSCRIPTS

“When No Enforceable Locus Exists: Availability and Representability in a Financial AI Agent Stack.”
Mulan Zhou, Yan Xue. Accepted — main conference research article, 7th ACM International Conference on AI in Finance (ICAIF 2026).

“IDV: Customer-Context-Aware Multi-Engine Architecture for AI-Agent Intent Drift Detection in Finance.”
Mulan Zhou, Yan Xue. 2026 IEEE Computational Intelligence in Financial Engineering and Economics (CIFEr); Best Paper Award. DOI: 10.1109/CIFER67845.2026.11692374. | SSRN preprint

RESEARCH ENGINEERING

Instinctive Network | AI Research & Product Development — 2025–Present

- Developing structured Intent Checkpoints and evaluation cases for authorization drift during multi-step agent execution, linking intent specifications to auditable decisions and impact assessment.
- Contributing to ConstraintFlow (open-source project in development): replay-based checks for dropped, truncated, or unenforceable constraints at agent instruction-loading and execution boundaries.
- Building pinned-version loader fixtures, reproducible failure records, and conservative release checks that distinguish verified constraint delivery from downstream behavioral compliance.

Gamegenie & Noobled Studios | Roblox Game Design & Development — 2024–2025

- Built server-side Lua systems for games with 1.7M+ cumulative visits, using engagement and click-through analysis to inform retention improvements.

TECHNICAL & RESEARCH SKILLS

Machine learning & model internals: Python, PyTorch, Hugging Face Transformers, scikit-learn; Qwen2.5-Instruct (1.5B/14B); GPU-based LLM experiments, residual-stream probes, linear readouts, activation interventions, NLI, sentence embeddings.

Experimental design: Preregistration, held-out and matched-control studies, structural interventions, causal falsification, failure-mode analysis, bootstrap inference, calibration; F1, AUROC/AUPRC, precision/recall, ECE; benchmark and annotation design.

Engineering & research tools: NumPy, pandas, Git/GitHub, Jupyter, LaTeX, Label Studio; Java, JavaScript, Lua, HTML/CSS; replay-based test harnesses.

AWARDS & LEADERSHIP

Best Paper Award — IEEE CIFEr 2026. Recognized for the IDV paper.

DECA Finance, California Championships (2023). 2nd place, Individual Finance; international conference qualifier.

UCSD Esports — Team Manager & Coach (2025–Present). Coordinated competition logistics and weekly review; coached 30+ players to top-64/128/256 finishes in North American championship play.